Prelab 0

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1 Basics of optical alignment

1.1 task 1.(a)

The standard for laser classification changed over time, the currently accepted version is one that specified by IEC 60825-1 standard

Class 1 : MPE cannot be exceeded when viewing with naked eyes or through any type of optics.

Class 1M : Safe in all conditions except through magnifying optics

Class 2 : Since blink reflex limits the exposure time to laser under 0.25 seconds [2]. It only apply to visible light laser. For continuous wave laser, the power is limited to 1mW.

Class 3R : MPE is exceeded but risk of injury remained low. For visible continuous wave,the power is limited to 5mW

Class 3B : Hazardous to eyes if exposed directly, its diffused reflection, however, is not For continuous wave with wavelength from 315nm to far infrared, the power is limited to 0.5w. For pulsed laser in visible light, the energy is limited to 30mJ.

Class 4 : All lasers exceeding the Class 3B AEL are Class 4 Lasers.

Optical Density : The measurement of how much light can protective optics attenuate.

MPE:the aximum amount of laser can a person take before suffering from any form of injury.

1.2 1.(b)

the document {laser+MPE...calculation} provide an equation that is said to be suitable for visible lasers in Example 1.1, where it mentioned a Table 3, which I can't find. According to the equation

$$MPE = 1.8t^{\frac{3}{4}} \times 10^{-3} J \cdot cm^{-2}$$

radiant exposure $H = 0.636mJ \cdot cm^{-2}$

in terms of irradiance,

$$\begin{aligned} E =& \frac{H}{t} \\ =& 2.55mW \cdot cm^{-2} \end{aligned} \tag{1}$$

1.3 1.(c)

according to the video, cornea, retina and the lens are most susceptible to damage.

1.4 1.(d)

according to the Wikipedia page given in the asignment, visible to near-infrared is the most dangerous to the eye

1.5 1.(e)

choose goggles according to the specific work mode, wave length range, maximum laser power and coresponding minimum portection level needed for the given power

1.6 task 2

Fresnel reflection occurs when laser encounter impedance discontinuity, i.e. entering lenses made of fused silica. The amount of laser reflected is discribed by reflecance, calculated by using the following equation:

$$R = (\frac{n_2 - n_1}{n_2 + n_1})^2$$

where n_1 and n_2 are the refractiveindex of the mediums int this task, $n_1 \approx 1$ and $n_2 \approx 1.458$ therefore

$$R = 0.042$$

this means 4.2% of the light is reflected while 95.8% of the light when in the fused silica lens.

Now consider the question when a ray of light went through the lens. the discontinuity of impedance occurs twice: both entering and exiting the lens. So the loss during the process can be calculated as such:

$$1 - (1 - R)^2$$

the result is: 9.75% of the light is lost going through the lens.

1.7 task 3

the power of the laser can be calulatd as such:

$$P = \frac{V_{load}}{R_{load}} R_{DET36A2}$$

according to the product page of DET36A2, in the conditions given in the question, $R_{DET36A2} \approx 0.3A/W$ [1], therefore the power of the laser is:

$$P \approx 12\mu W$$

2 Gaussian Beams and Telescopes

2.1 task 4

The ABCD matrix of the objective lens, the gap between lenses and the eyepiece lens are given as follows:

$$M_{objective} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_o} & 1 \end{bmatrix}$$
$$M_{gap} = \begin{bmatrix} 1 & f_o + f_e \\ 0 & 1 \end{bmatrix}$$
$$M_{eyepiece} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_e} & 1 \end{bmatrix}$$

the ABCD matrix of the telescope can be given by:

$$\begin{aligned} M_{telescope} &= \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_e} & 1 \end{bmatrix} \begin{bmatrix} 1 & f_o + f_e \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_o} & 1 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{f_e}{f_o} & f_o + f_e \\ 0 & -\frac{f_o}{f_e} \end{bmatrix} \end{aligned} \tag{2}$$

The vector of the input light ray that we actually care:

$$\begin{pmatrix} w \\ 0 \end{pmatrix}$$

and the output vector is

$$\begin{pmatrix} -\frac{f_e}{f_o} w \\ 0 \end{pmatrix}$$

it is clear that the waist of output ray is $\frac{f_e}{f_o} w$, to answer the question, the ratio of input beam waist to output beam wasit is $\frac{f_o}{f_e}$.

2.2 task 5

I am finding this question a little strange, maybe I'm not understanding it corectly.

When we are looking into the telescope, our eyes are capturing the light coming out of the eyepiece lens, which is parallel, as stated in the question. When we are looking at the world using our naked eyes, our eyes are also capturing parallel light rays. We can see images with our naked eyes, so it is natural for our eyes to be able to see images from the telescope, because they are taking in input of the same nature.

Simply put: naked eyes can translate parallel light rays into image, the light rays coming out of telescope are parallel, therefore we can see image by looking into a telescope

As for why naked eyes can translate parallel light rays into image, it is because the optical lens in our eyeballs can focus parallel rays of light onto our retina, where light wave of certain wavelength is translated into some sort of signal apprehensible by human brain.

2.3 task 6

The magnification M of the telescope is given by the ratio of the output ray angle to the input ray angle. We can easily derive from the ABCD matrix obtained in task 4 that the absolute value of the ratio of the output to input angles is $\frac{f_o}{f_e}$ and that is the magnification

2.4 task 7

the modification I'd make is as the picture depicted below:

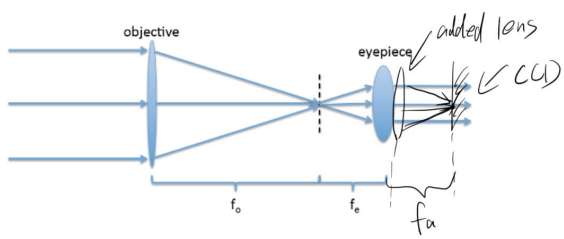


Figure 1: modification that allow CCD to capture the image

the ABCD matrix of the system with the added lens is

$$\begin{bmatrix} 1 & f_a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_a} & 1 \end{bmatrix} \begin{bmatrix} \frac{f_e}{f_o} & f_o + f_e \\ 0 & -\frac{f_o}{f_e} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -\frac{f_o}{f_e f_a} \\ -\frac{f_e}{f_o f_a} & -\frac{1}{f_a}(f_o + f_e) - \frac{f_o}{f_e} \end{bmatrix}$$

$$\begin{bmatrix} 0 & -\frac{f_o}{f_e f_a} \\ -\frac{f_e}{f_o f_a} & -\frac{1}{f_a}(f_o + f_e) - \frac{f_o}{f_e} \end{bmatrix} \begin{bmatrix} r_1 \\ r_1' \end{bmatrix} = \begin{bmatrix} \cdots \\ -\frac{f_e r_1}{f_o f_a} - r_1' \left(\frac{f_o + f_e}{f_a} + \frac{f_a}{f_e} \right) \end{bmatrix}$$

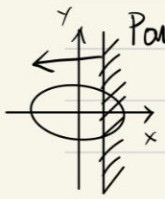
Figure 2: modification that allow CCD to capture the image

the magnification changed after my modification

2.5 task 8

8. $I(x,y) = \frac{2P}{\pi w_x w_y} e^{-\frac{2}{w_x^2}x^2} e^{-\frac{2}{w_y^2}y^2}$

$$\mathrm{erf} = \frac{2}{\sqrt{\pi}} \int_{-\infty}^{x} e^{-t^2} dt$$



Power measured: $P(x) = \frac{2P}{\pi w_x w_y} \int_{-\infty}^{+\infty} dy \int_{-\infty}^{x} dt\, e^{-\frac{2}{w_x^2}t^2} e^{-\frac{2}{w_y^2}y^2}$

1. $\int_0^x e^{-\frac{2}{w_x^2}t^2} dt = \frac{w_x}{\sqrt{2}} \int_0^x e^{-(\frac{\sqrt{2}}{w_x}t)^2} d\frac{\sqrt{2}}{w_x}t$, Let $\frac{\sqrt{2}}{w_x}t = a$, we have

$$\frac{w_x}{\sqrt{2}} \int_0^x e^{-(\frac{\sqrt{2}}{w_x}t)^2} d\frac{\sqrt{2}}{w_x}t = \frac{w_x}{\sqrt{2}} \int_0^{\frac{\sqrt{2}}{w_x}x} e^{-a^2} da = \frac{w_x}{\sqrt{2}} \cdot \frac{\sqrt{\pi}}{2} \mathrm{erf}\left(\frac{\sqrt{2}}{w_x}x\right)$$

2. $\int_{-\infty}^{0} e^{-\frac{2}{w_x^2}t^2} dt = \cdots = \frac{w_x}{\sqrt{2}} \cdot \frac{\sqrt{\pi}}{2} \mathrm{erf}(\infty) = \frac{w_x}{\sqrt{2}} \cdot \frac{\sqrt{\pi}}{2}$

3. $\int_{-\infty}^{+\infty} e^{-\frac{2}{w_y^2}y^2} dy = \sqrt{\frac{\pi}{\frac{2}{w_y^2}}} = w_y \sqrt{\frac{\pi}{2}}$

$$\therefore P(x) = \frac{2P}{\pi w_x w_y} \cdot \frac{w_x}{\sqrt{2}} \cdot \frac{\sqrt{\pi}}{2} \left(1 + \mathrm{erf}\left(\frac{\sqrt{2}}{w_x}x\right)\right) w_y \sqrt{\frac{\pi}{2}}$$

$$= \frac{P}{2} \left(1 + \mathrm{erf}\left(\frac{\sqrt{2}}{w_x}x\right)\right)$$

Figure 3: modification that allow CCD to capture the image

it is possible to determine the waist by taking measurement with different xs then perform non-linear fit.

this is slightly different from the equation given in the problem, but if you look closely, the only difference is the direction of the knife edge. Mine is going from +inf. to -inf., the equation in the problem is going the opposite direction.

as for w_y, the x and y in this problem are essentially the same. You can substitute all of the x in the equation with y, and then it is the equation for w_y

2.6 task 9

I wrote a little program that dosen't satisfy all of the function mentioned in Lab II, but is capable to give answer to this question. I'll complete the program later.

an example of running this program:

```
    parameters:
    R_in = np.inf
    waist_in = 1e-3
```

```
    WaveLength = 633e-9
    f_e = 1
    f_o = 1

    retsult:
    the curvature radius: -14.315791550416103 m
    the waist: 0.0010781433740509 m
```

this is my program in text:

```python
import numpy as np

def Gaussian_Beams(R, WaveLength, waist):
    """Gaussian Beam discribed by 1/q, returns 1/q"""
    return 1/R - 1j*WaveLength/(np.pi*waist**2)

def Transformation(q_r, TransformMatrix):
    """How the reciprocal of q evolve along the way, returns 1/q' """
    return (TransformMatrix[1,0]+TransformMatrix[1,1]*q_r)/(TransformMatrix
        [0,0]+TransformMatrix[0,1]*q_r)

def Radius(q_r):
    """Calculate curvature radius, returns curvature radius"""
    return 1 / np.real(q_r)

def Waist(q_r, WaveLength):
    """Calculate waist, returns waist"""
    return np.sqrt(-1*WaveLength/(np.imag(q_r)*np.pi))

# set parameters
R_in = np.inf
waist_in = 1e-3
WaveLength = 633e-9
f_e = 1
f_o = 1
TransformMatrix = np.array([[-1*f_e/f_o, f_e + f_o],
                            [     0,     -1*f_o/f_e]])

# calculate initial q in reciprocal
q_r_in = Gaussian_Beams(R_in, WaveLength, waist_in)

# How the reciprocal of q evolve along the way
q_r_out = Transformation(q_r_in, TransformMatrix)

# result radius
R_out = Radius(q_r_out)

# result waist
waist_out = Waist(q_r_out, WaveLength)

print(" the curvature radius:", R_out,"m\n", "the waist:" , waist_out, "m")
```

References

[1] thorlabs. Free-space biased detectors, N/A. [Online; accessed 28-September-2024].

[2] Wikipedia contributors. Laser safety — Wikipedia, the free encyclopedia, 2024. [Online; accessed 28-September-2024].